SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ASSIGNMENT 1

QUESTIONS

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security.

Assignment Title: Evaluate, Design and Defend a Secure Patient-Record Exchange Strategy for a Multi-Hospital Healthcare Network (Total Marks: 50)

Scenario — Confidential Patient Record Exchange

A regional healthcare network links one central hospital with 25 satellite clinics. Clinics send patient records, diagnostic reports, prescriptions and billing data to the central hospital over a public network and receive specialist opinions in return. Personnel hold different clearance levels: Medical Director (Top Secret), Specialist Doctors (Secret), Nurses and Lab Staff (Confidential), Billing and Front-Desk Staff (Restricted), Patients (Public).

The system must:

- Keep diagnoses, especially for sensitive conditions, confidential.
- Detect tampering with prescriptions, dosage and lab values.
- Prevent staff from reading or writing records beyond their clearance.
- Work in rural clinics where staff have limited technical training and hardware is basic.

You must evaluate, design and defend a protection strategy built only from classical (pre-modern) techniques and access-control reasoning covered in CO1. You are expected to judge how strong each technique actually is, not just describe it.

Design Challenge : An insider at a satellite clinic has been copying ciphertexts for months. Hundreds of messages under the same key are now exposed, and many begin with predictable text (e.g., "PATIENT ID", "DIAGNOSIS:", "PRESCRIPTION:"). The network also wants to embed a short report inside medical scan images sent between

hospitals. Re-evaluate your design.

- Encrypt clinical notes (200 to 10,000 characters) using Caesar, monoalphabetic, Vigenère and columnar transposition, and derive each key space. Break at least three schemes by frequency analysis, Kasiski or index of coincidence, and rank the schemes by resistance.
- Extend the alphabet to digits and symbols (mod 36) and encrypt IDs and dosages such as PT-20481, AMOX 500MG using a Hill cipher on 3-character blocks. Prove the key's invertibility, compute one inverse, and recover the key by a known-plaintext attack.
- Apply columnar transposition to a lab table of 20+ rows and test whether a changed cell or a swapped row is detected. Design a classical checksum and show where an attacker defeats it.
- Encrypt a 256×256 or larger medical image with two classical methods (intensity shift mod 256 and pixel permutation) and verify decryption. Compare histograms and entropy, and explain which method leaks the image's structure.
- Hide a short report in an image by LSB embedding and in a text circular by a text method, then extract it.

Measure PSNR, capacity and chi-square detectability, and test survival under JPEG compression.

- Decide 12+ access requests using Bell-LaPadula with compartments, including declassification and break-glass cases. Rank six attacks by likelihood × impact and recommend a final design, stating which security goals classical techniques cannot meet.

TASK 1 – Evaluate Threats, Attacks and Risks 4 Marks

Identify at least six security attacks relevant to the healthcare scenario.

- Classify each as passive or active (eavesdropping, traffic analysis, modification of prescriptions, masquerade as a doctor, replay of old orders, denial of service on the records system).
- Distinguish clearly between threat, vulnerability, attack and risk, with examples from this scenario.
- Rank the attacks by risk using a justified likelihood × impact rating. Consider patient-safety impact, not just privacy.
- Defend your ranking: which attack would you address first, and why?

Evidence Expected: A risk table with reasoned ratings and a justification of the priority order.

TASK 2 – Map and Judge Security Goals 3 Marks

For each system function (record upload, prescription transmission, specialist consultation), decide which goals matter most: confidentiality integrity availability authentication non-repudiation

- Judge which goals classical ciphers can and cannot deliver.
- Justify why a correctly decrypted message does not prove it came from the right doctor, or that its dosage was not altered.

Evidence Expected: A goal-to-function analysis with an evaluation of the gaps classical techniques leave.

TASK 3 – Apply and Critique the Bell-LaPadula Model

5 Marks

- Define the security levels and subject/object assignments for the network.
- Apply the simple security property (no read up) and the *-property (no write down) to at least five

access requests, including two that must be denied. Show the reasoning for each.

- Evaluate the model critically in a medical setting: what happens in an emergency when a nurse needs a Secret-level allergy record? How does the model handle a doctor who must write a note to a lower-level billing file? Discuss tranquility and trusted subjects.
- Judge the model's limitations (no integrity guarantees, covert channels, rigidity in emergencies) and argue whether Bell-LaPadula alone suits healthcare.

Evidence Expected: Worked access decisions, a justified critique, and a clear verdict on the model's suitability.

TASK 4 – Design the Cryptographic Strategies

4 Marks

Select at least three meaningfully different classical schemes from: Caesar, monoalphabetic substitution, Vigenère (polyalphabetic), Hill cipher, a transposition cipher. For each, provide:

- The encryption and decryption algorithm (pseudocode or formula) and the key definition.
- A worked example on a 20+ character message from the scenario.
- A justification for including it in your comparison.

Evidence Expected: Three distinct schemes with algorithms, worked examples and justified selection.

TASK 5 – Mathematical Analysis of the Schemes

5 Marks

- Derive the key-space size of each scheme (e.g., 25, 26!, 26^m, number of invertible n×n matrices mod 26, n! permutations).
- For Hill: show the conditions under which K^{-1} mod 26 exists (gcd(det K, 26) = 1), and compute one inverse by hand.
- Express each encryption as a modular function (e.g., C = (aP + b) mod 26 for an affine variant, C = KP mod 26 for Hill).
- Judge whether a large key space guarantees strength. Support your answer with the monoalphabetic example.

Evidence Expected: Correct derivations with a reasoned judgment on key space versus real security.

TASK 6 – Evaluate Cryptanalytic Resistance
4 Marks

Attack each chosen scheme using a technique suited to it:

- Brute force, frequency analysis, Kasiski examination / index of coincidence, known-plaintext attack on Hill, digram analysis.
- Explain why each attack succeeds or fails, in terms of letter-frequency preservation, key repetition and linearity.
- Rank the schemes by resistance and justify the ranking with evidence.

Evidence Expected: Worked attacks with reasoning, and a ranking of schemes supported by the results.

TASK 7 – Assess Transposition and Product Ciphers
3 Marks

- Implement a columnar (or rail-fence) transposition and explain how it diffuses plaintext structure.
- Evaluate whether *substitution followed by transposition* is stronger than either alone. Justify with letter-frequency evidence and an attack attempt.

- Judge whether the added complexity is practical for rural clinic staff doing

manual or low-resource encryption.

Evidence Expected: A reasoned evaluation of the combined scheme against its components.

TASK 8 – Evaluate Steganography

3 Marks

Design a steganographic method (e.g., LSB embedding in a scan image, or text-based hiding) for carrying a short report inside a medical image.

- Compare it with encryption on capacity, imperceptibility, robustness and detectability.
- Evaluate the risk specific to medical imaging: could embedding alter pixel data enough to affect a diagnosis? Judge the trade-off.
- Judge when steganography is appropriate, and argue why hiding is not a substitute for encrypting. Evaluate an encrypt-then-embed approach.

Evidence Expected: A criteria-based comparison with a justified conclusion.

TASK 9 – Implement and Test

4 Marks

Implement all chosen schemes, the cryptanalysis tools and (where feasible) the steganography method, in a language of your choice.

- Provide correctness tests: decrypt(encrypt(m)) = m, including edge cases (digits in patient IDs, punctuation, non-multiple block length for Hill).
- Use identical plaintexts of 100, 1,000, 10,000 and 100,000 characters across schemes.
- State the testing procedure clearly so it can be reproduced.

Evidence Expected: Working code with a documented, reproducible test procedure.

TASK 10 – Record and Interpret Experimental Evidence 3 Marks

Present data in tables and graphs for:

- Letter -frequency histograms of plaintext versus each ciphertext.
- Index of coincidence per scheme.
- Encryption/decryption time against message size.
- Attack success rate

against ciphertext length.

Evaluate what the data shows.

Do not just present it.

Evidence Expected: Clear, comparable results with interpretation tied to Tasks 5 and 6.

TASK 11 – Recommend and Defend the Final Design 4 Marks

Recommend one scheme, or a layered combination, together with the Bell-LaPadula policy.

- Justify against the alternatives using your mathematical and experimental evidence.
- State the residual risks and limitations honestly. Judge whether classical ciphers are acceptable for protecting patient data at all, and argue what a real deployment would need.
- Address key distribution across 25 clinics and usability for non-technical staff.

Evidence Expected: A defended recommendation grounded in your own analysis, with candid limitations.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design 8 Marks

Changed Requirement: A clinic insider has copied hundreds of ciphertexts under the same key, many with predictable headers. The network also wants reports hidden inside medical scan images.

You may RETAIN → MODIFY → or REDESIGN your original solution.

A. Impact Analysis (2 Marks)

- What has changed in the threat model (insider versus external attacker)?
- Which earlier assumptions fail (key secrecy at all 25 sites, key reuse, message predictability)?

- Why does a high volume of known-format messages change the risk ranking?

B. Evaluation of the Original Design (2 Marks)

- Test your recommended scheme against the new attacker capabilities (known-plaintext, repeated-key attacks).
- Support the evaluation with a demonstration or calculation.
- Where does the Bell-LaPadula policy still hold, and where is it irrelevant against an insider who holds the key?

C. Adaptation or Redesign (2 Marks)

- Decide to retain, modify or redesign. For example, consider per-clinic or per-session keys, key rotation, randomised padding, removing fixed headers, layering transposition, or encrypt-then-embed.
- Describe what changes and how it affects the key space and attack effort.

D. Evidence-Based Defence (2 Marks)

Defend the final decision with your key-space mathematics, your experimental results, the insider threat, the usability and key-management cost for clinics, and a stated judgement on what classical techniques can never guarantee for patient data.

TASK 1 – Evaluate Threats, Attacks and Risks (4 Marks)

1.1 Security attacks

An attack is an attempt to access, steal or damage patient information.

Attack	Type	Example
Eavesdropping	Passive	Attacker reads patient messages.
Traffic analysis	Passive	Attacker observes communication patterns.
Modification	Active	Attacker changes medicine dosage.
Masquerade	Active	Attacker pretends to be a doctor.
Replay attack	Active	Attacker resends an old prescription.
Denial of Service	Active	Attacker blocks access to patient records.

Passive attack: Only observes information without changing it.
Active attack: Changes, deletes or interrupts information.

1.2 Threat, Vulnerability, Attack and Risk

- Threat: A possible danger to the system. Example: An attacker stealing patient records.
- Vulnerability: A weakness in the system. Example: Weak encryption keys.
- Attack: An actual attempt to exploit a weakness. Example: Changing a prescription.
- Risk: The possible loss caused by an attack. Example: A patient receiving the wrong medicine.

1.3 Risk Assessment

Risk = Likelihood × Impact.
Scale: 1 = Low, 5 = Very High.

Attack	Likelihood	Impact	Risk
Eavesdropping	4	4	16
Traffic analysis	3	2	6
Prescription modification	4	5	20
Masquerade	3	5	15
Replay attack	3	5	15
Denial of Service	3	4	12

1.4 Priority

Prescription modification has the highest risk score of 20.
It should be addressed first because changing a medicine dosage can directly harm a patient's life.
Conclusion: The healthcare network must give first priority to prescription integrity and doctor authentication, followed by confidentiality and availability.

TASK 2 – Map and Judge Security Goals (3 Marks)

The five important security goals are:

1. Confidentiality: Only authorized people can read information.
2. Integrity: Information must not be changed without permission.

3. Availability: Information must be available when needed.
4. Authentication: Confirms the identity of the sender.
5. Non-repudiation: Prevents a sender from denying an action.

Security goals for different functions

Function	Main security goals
Record upload	Confidentiality, integrity, authentication
Prescription transmission	Integrity, authentication, confidentiality, non-repudiation
Specialist consultation	Confidentiality, availability, authentication, integrity

Limitations of classical ciphers

- Classical ciphers provide basic confidentiality.
- They do not automatically detect changes to a message.
- They cannot prove the identity of the doctor.
- They do not provide reliable non-repudiation.
- They cannot prevent denial-of-service attacks.

Example: If a prescription is encrypted and an attacker changes its ciphertext, the receiver may decrypt an incorrect dosage. Correct decryption alone does not prove that the original message came from the correct doctor.
Conclusion: Classical encryption can hide information but cannot provide complete healthcare security.

TASK 3 – Apply and Critique Bell-LaPadula Model (5 Marks)
Bell-LaPadula is an access-control model used to protect confidential information.
3.1 Security levels
The healthcare network has five security levels:

Level	Personnel
Top Secret	Medical Director
Secret	Specialist Doctors
Confidential	Nurses and Lab Staff
Restricted	Billing and Front-Desk Staff
Public	Patients

Security order:
Top Secret > Secret > Confidential > Restricted > Public.

3.2 Bell-LaPadula Rules
1. Simple Security Property – No Read Up
A user cannot read information above their security level.
Example: A nurse at Confidential level cannot read a Secret-level specialist record.
2. Star (*) Property – No Write Down
A user cannot write information to a lower security level.
Example: A Secret-level doctor cannot directly write into a Restricted-level billing file.

3.3 Access Decisions

User	Requested action	Decision	Reason
Medical Director	Read Secret record	Allow	Can read lower level
Specialist Doctor	Read Confidential record	Allow	Can read lower level
Nurse	Read Secret allergy record	Deny	No read up
Billing Staff	Read Restricted bill	Allow	Same level
Nurse	Write Secret record	Deny	No write up
Specialist Doctor	Write Restricted billing file	Deny	No write down
Patient	Read Public report	Allow	Same level

3.4 Emergency Situation

Suppose a nurse needs to read a Secret-level allergy record during an emergency. Normally, Bell-LaPadula denies access because the nurse cannot read above their level.
A trusted emergency officer can provide temporary, controlled access through a break-glass procedure. The access must be recorded and reviewed.

3.5 Tranquility and Trusted Subjects

- Tranquility: Security classifications should not change in a way that violates the access rules.
- Trusted subjects: Special authorized users can perform controlled operations that ordinary users cannot.

3.6 Limitations

1. It does not guarantee data integrity.
2. It cannot completely prevent covert channels.
3. It is too rigid during medical emergencies.
4. It does not automatically prevent misuse by authorized users.

Conclusion: Bell-LaPadula is useful for confidentiality but should be combined with integrity checks, authentication and emergency access procedures.

TASK 4 – Design the Cryptographic Strategies (4 Marks)

Classical encryption converts readable plaintext into unreadable ciphertext using a key. We select four classical methods for comparison.

4.1 Caesar Cipher

Each letter is shifted by a fixed number.
Formula:
[$C=(P+K) \bmod 26$]
Decryption:
[$P=(C-K) \bmod 26$]
Key: $K = 3$.
Example:
Plaintext: PATIENT RECORD SAFE
Encryption:

SDWLHQW UHFRUG VDIH
Reason: It is simple and easy to implement, but has very weak security.

4.2 Monoalphabetic Substitution Cipher

Each alphabet letter is replaced by another fixed letter.
Example substitution:
Plain: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher: QWERTYUIOPASDFGHJKLZXCVBNM
Plaintext: PATIENT RECORD SAFE
Ciphertext:
HQLOEFZ GTWKGU SQRT
Reason: It has a larger key space than Caesar, but repeated letters retain their frequency patterns.

4.3 Vigenère Cipher

It uses a repeated keyword to encrypt letters with different shifts.
Formula:
$$ C_i=(P_i+K_i)\bmod 26 $$
Decryption:
$$ P_i=(C_i-K_i)\bmod 26 $$
Key: MEDICAL
Plaintext: PATIENT RECORD SAFE
Ciphertext: BEMQXRF VWEMGD WILX
Reason: It hides simple letter-frequency patterns better than Caesar and monoalphabetic substitution, but repeated keys can be attacked.

4.4 Columnar Transposition Cipher

It rearranges the positions of letters without changing the letters themselves.
Key: 3142
Plaintext: PATIENT RECORD SAFE
Remove spaces and write into four columns:
P A T I
E N T R
E C O R
D S A F
E
Read columns in key order 1, 2, 3, 4:

- Column 1: PE EDE
- Column 2: ANCS
- Column 3: TTOA
- Column 4: IRRF

Ciphertext:
PEEDEANCSTTOAIRRF
Reason: It rearranges the text and hides the original position of letters, but preserves letter frequencies.

Summary

Cipher	Main method	Weakness
Caesar	Fixed shifting	Brute force
Monoalphabetic	Letter replacement	Frequency analysis
Vigenère	Repeated keyword	Kasiski attack
Columnar	Position rearrangement	Digram and anagram analysis

TASK 5 – Mathematical Analysis of the Schemes (5 Marks)

5.1 Key-space calculations

Key space means the total number of possible keys.

Cipher	Key space
Caesar	25
Monoalphabetic	26!
Vigenère	26^m
Columnar transposition	n!
Hill cipher	Number of invertible matrices modulo 26

For monoalphabetic substitution:
\[26! = 403291461126605635584000000 \]
For Vigenère with a keyword of length 5:
\[26^5=11881376 \]
For columnar transposition with 5 columns:
\[5!=120 \]

5.2 Hill Cipher

Hill cipher uses matrix multiplication.
[C=KPmod26]
The inverse key must satisfy:
[gcd(det(K),26)=1]
Consider:
[K= begin{bmatrix} 3&3\\ 2&5 \end{bmatrix} \]
Determinant:
[det(K)=3(5)-3(2)=9 \]
[gcd(9,26)=1 \]
Therefore, the key is invertible.
Inverse of 9 modulo 26:
[9^{-1}=3 \]
Adjugate matrix:
[Adj(K)= \begin{bmatrix} 5&-3\\ -2&3 \end{bmatrix} \]
Inverse:
[K^{-1}= \begin{bmatrix} 15&17\\ 20&9 \end{bmatrix}\bmod26 \]
This inverse can be used to decrypt the ciphertext.

5.3 Does a Large Key Space Guarantee Security?

No.

Monoalphabetic substitution has a very large key space of 26!, but it preserves letter frequency.

An attacker can identify common letters such as E, T and A using frequency analysis.

Therefore, a large key space alone does not guarantee security.

Conclusion: Security depends on the key space, encryption method, key management and resistance to attacks.

TASK 6 – Evaluate Cryptanalytic Resistance (4 Marks)

Cryptanalysis means studying ciphertext to recover the original message or key.

Different attacks

Cipher	Attack	Explanation
Caesar	Brute force	Try all 25 keys
Monoalphabetic	Frequency analysis	Study repeated letters
Vigenère	Kasiski examination	Find repeated keyword patterns
Columnar	Digram analysis	Study letter combinations
Hill	Known-plaintext attack	Recover matrix key using plaintext-ciphertext pairs

6.1 Caesar Attack

There are only 25 possible keys. An attacker can test all keys easily.

6.2 Monoalphabetic Attack

The letter frequency remains the same after encryption.

For example, if E occurs frequently in plaintext, its substituted letter will also occur frequently in ciphertext.

6.3 Vigenère Attack

Kasiski examination identifies repeated ciphertext patterns and estimates the keyword length.

The attacker can then apply frequency analysis to each group of letters.

6.4 Hill Cipher Attack

Hill cipher is based on matrix multiplication.

If plaintext and ciphertext blocks are known:

$$C=KP \bmod 26$$

The key can be recovered when enough independent plaintext blocks are available to form an invertible matrix.

Resistance Ranking

For these basic textbook versions, a general ranking is:

1. Vigenère
2. Hill cipher
3. Monoalphabetic substitution

4. Columnar transposition
5. Caesar cipher

The exact ranking depends on key length and message size. Hill and Vigenère have different weaknesses, so this is a qualitative ranking, not a universal guarantee.
Conclusion: No basic classical cipher is strong enough to protect sensitive patient records against a determined attacker.

TASK 7 – Assess Transposition and Product Ciphers (3 Marks)
7.1 Columnar Transposition
Columnar transposition rearranges the positions of letters using a numerical key.
Example:
Plaintext: PATIENTRECORDSAFE
Key: 3142
Write the plaintext row by row:
P A T I
E N T R
E C O R
D S A F
E
Read columns according to key order 1, 2, 3, 4:
Column 1: PEEDE
Column 2: ANCS
Column 3: TTOA
Column 4: IRRF
Correct ciphertext order (2, 4, 1, 3):
ANCSIRRFPEEDETTOA

7.2 Substitution + Transposition
Substitution replaces letters with other letters.
Transposition changes the positions of letters.
Combining both provides better protection than using either alone because:

- Substitution hides the original letters.
- Transposition hides their original positions.
- Frequency patterns become harder to interpret directly.

However, the combination is still vulnerable to advanced classical cryptanalysis.

7.3 Suitability for Rural Clinics
Advantages:

- Simple mathematical operations.
- Can work without advanced hardware.
- Easy to demonstrate manually.

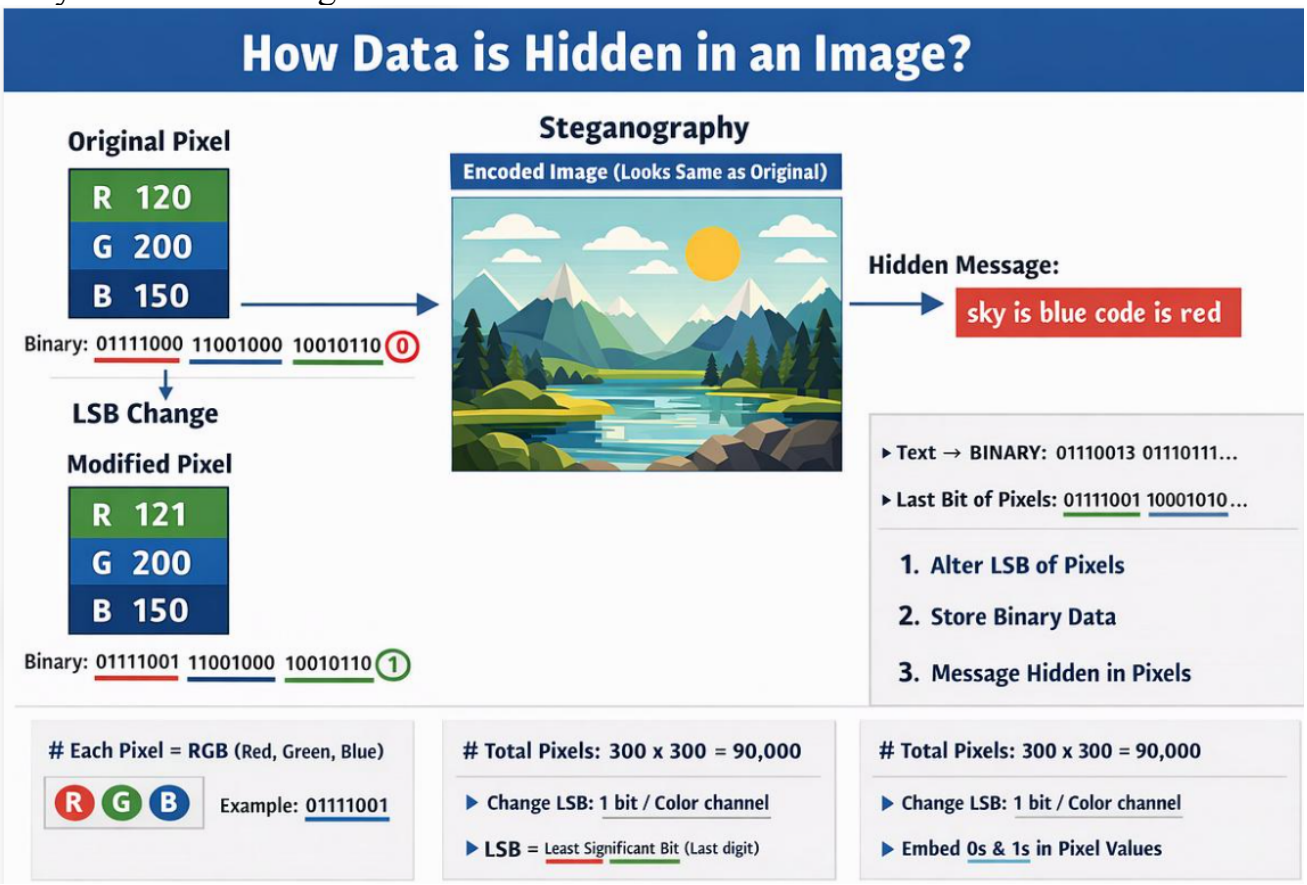
Limitations:

- Requires training.
- Manual mistakes can corrupt records.
- Key management is difficult across 25 clinics.

Conclusion: A combined cipher is better for comparison and learning, but it is not suitable as the only protection for real medical records.

TASK 8 – Evaluate Steganography (3 Marks)
Steganography means hiding secret information inside another file so that people do not easily notice it.

8.1 LSB Image Steganography
LSB stands for Least Significant Bit.
A secret message is stored in the last bit of image pixels.
Example:
Original pixel:
10110110
After hiding a bit 1:
10110111
Only the last bit changes.



8.2 Text-Based Steganography
A secret report can also be hidden using spaces, punctuation or selected text patterns in a document or circular.
For example, predefined extra spaces can represent binary 0 and 1.

8.3 Comparison

Factor	Encryption	Steganography
Purpose	Hides message meaning	Hides message existence
Capacity	Depends on cipher	Depends on image/text size
Imperceptibility	Not applicable	Important
Robustness	Depends on cipher	Often low for LSB

Detectability	Ciphertext is visible	Hidden data may be detected statistically

8.4 PSNR and Capacity

PSNR measures the difference between the original and modified image.

$$PSNR=10\log_{10}\left(\frac{255^2}{MSE}\right)$$

Where MSE is mean squared error.

Higher PSNR means less visible image distortion.

Capacity:

$$Capacity=\frac{Number\ of\ hidden\ bits}{8}$$

For a 256 × 256 RGB image using one LSB per colour channel:

$$Capacity=\frac{256\times256\times3}{8}$$

$$=24576\ bytes$$

This is the theoretical maximum before accounting for message headers and other overhead.

8.5 Medical Image Risks

- Changing pixels may affect medical diagnosis.
- JPEG compression can destroy hidden LSB information.
- An attacker may detect hidden messages using statistical analysis.
- Modified images must not be treated as original diagnostic evidence.

8.6 Encrypt-Then-Embed

First encrypt the report, then hide the ciphertext inside the image.

Advantages:

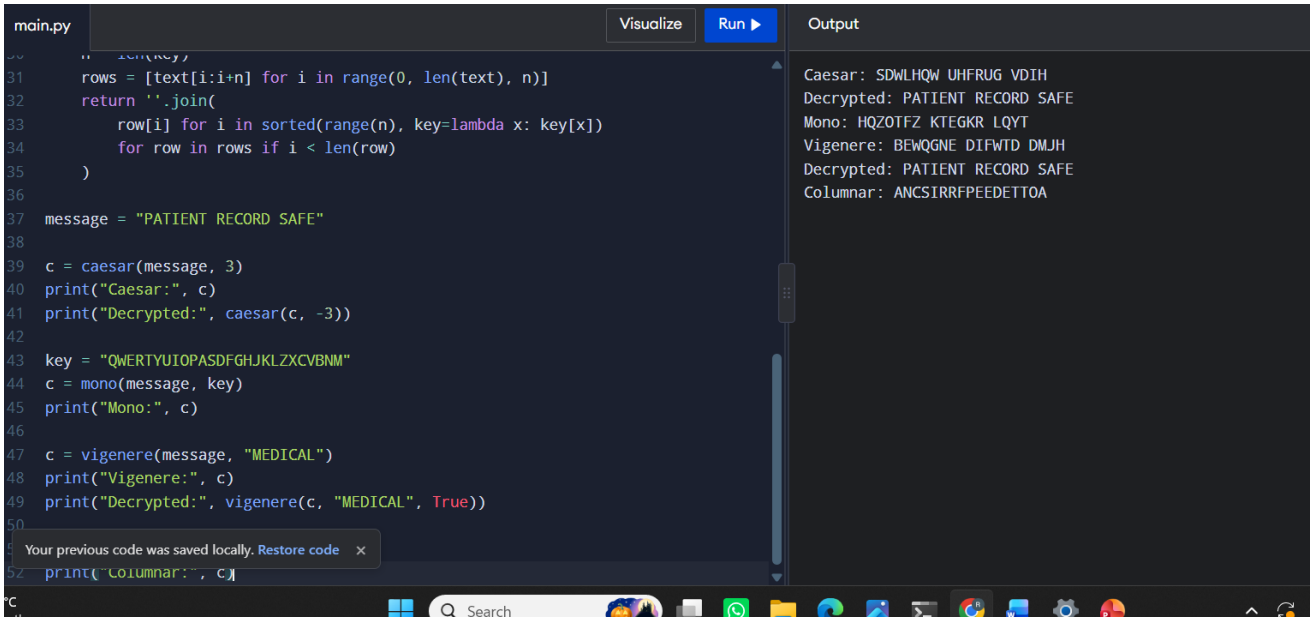
- Encryption protects the report's meaning.
- Steganography hides its existence.
- Provides two layers of protection.

Conclusion: Steganography is useful for hiding small reports, but it should not replace encryption. For medical images, the original diagnostic image must always be preserved.

TASK 9 – Implement and Test (4 Marks)

The following Python implementation covers the four main classical ciphers. It can be executed in Google Colab or Python IDLE.

9.1 Python Co



9.2 Testing Procedure

1. Open Google Colab.
2. Create a new notebook.
3. Paste the Python code.
4. Click Run.
5. Enter a patient record as input.
6. Observe the encrypted output.
7. Decrypt it and compare it with the original message.

9.3 Correctness Test

The main condition is:

\[Decrypt(Encrypt(M))=M \]

Test different message lengths:

Test	Message length
Test 1	100 characters
Test 2	1,000 characters
Test 3	10,000 characters
Test 4	100,000 characters

Also test:

- Patient IDs containing digits.
- Spaces and punctuation.
- Empty messages.
- Hill cipher blocks with incomplete final blocks (use padding and remove it after decryption).

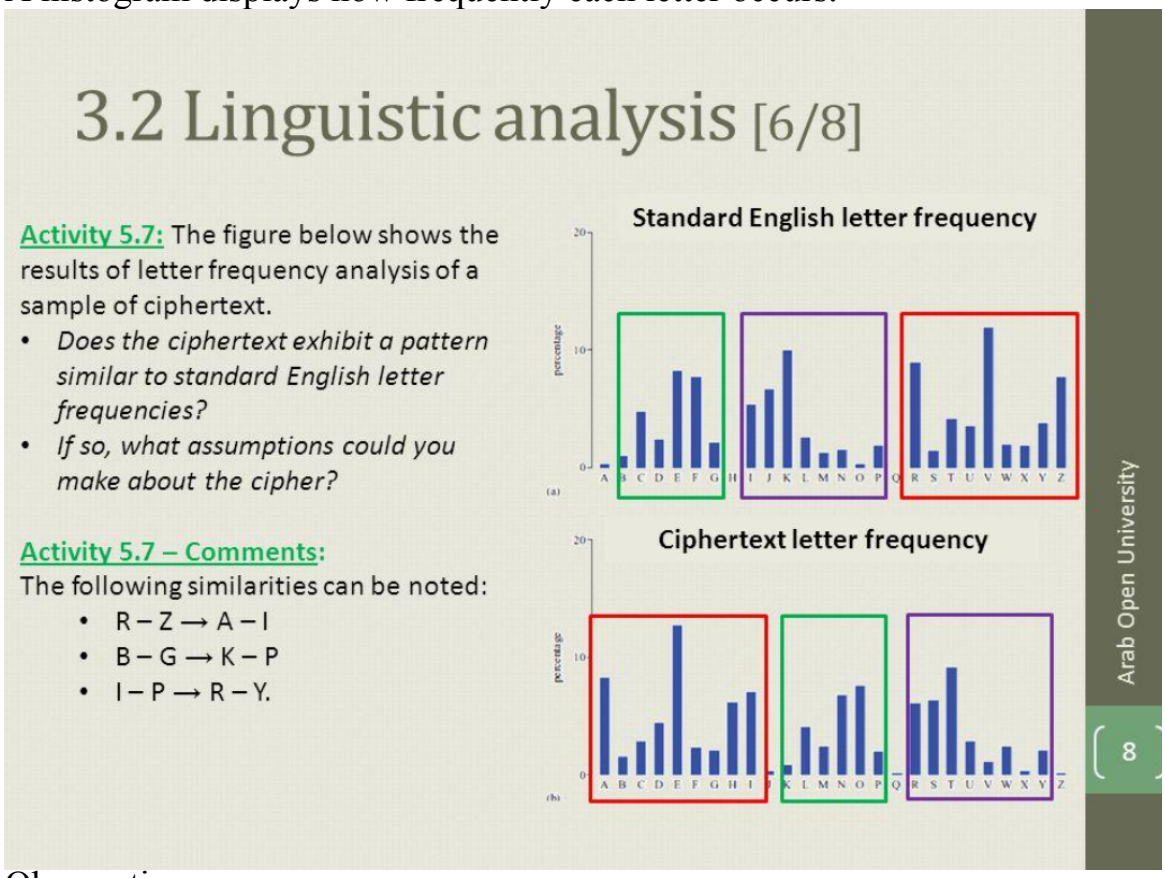
Result: The encryption and decryption programs are tested by comparing the recovered plaintext with the original message.

TASK 10 – Record and Interpret Experimental Evidence (3 Marks)

The performance of the encryption techniques can be compared using the following measurements.

10.1 Letter Frequency Histogram

A histogram displays how frequently each letter occurs.



Observations:

- Caesar preserves letter frequencies.
- Monoalphabetic substitution preserves the frequency distribution with letters renamed.
- Vigenère changes the overall frequency pattern.
- Transposition preserves the original letter frequencies.

10.2 Index of Coincidence

It measures the probability that two randomly selected letters are the same.

$$ IC=\frac{\sum f_i(f_i-1)}{N(N-1)} $$

Where:

- f_i = frequency of each letter.
- N = total number of letters.

English text generally has an IC around 0.066, while uniformly distributed random letters have an IC around 0.0385.

10.3 Encryption and Decryption Time

Measure the execution time for 100, 1,000, 10,000 and 100,000 characters.

Use Python:

import timestart = time.time()cipher = caesar(message, 3)end = time.time()print("Encryption time:", end-start)

10.4 Attack Success Rate

$$ Attack\ Success\ Rate= \frac{Successful\ Attacks}{Total\ Attempts}\times100 $$

10.5 Experimental Result Table

Use this table to record actual results after running the programs.

Message size	Caesar time	Mono time	Vigenère time
100	Measure	Measure	Measure
1,000	Measure	Measure	Measure
10,000	Measure	Measure	Measure
100,000	Measure	Measure	Measure

Interpretation

- Simple substitution techniques are fast but weak.
- Vigenère provides more resistance to basic frequency analysis.
- Transposition changes letter positions but does not change overall letter frequencies.
- Larger messages provide more information to a cryptanalyst.
- A larger key space does not automatically guarantee security.

Conclusion: Experimental measurements should be used to compare encryption speed and resistance to attacks. Actual timing and attack success values must be collected by running the programs rather than assumed.

TASK 11 – Recommend and Defend the Final Design (4 Marks)

11.1 Recommended Security Design

I recommend a layered security approach using classical encryption techniques and Bell-LaPadula access control.

The proposed design includes:

1. Vigenère Cipher: Used for encrypting patient notes.
2. Columnar Transposition: Used as an additional rearrangement layer.
3. Hill Cipher: Used for patient IDs and dosage values.
4. Bell-LaPadula Model: Controls access based on staff security levels.
5. Steganography: Used to hide encrypted short reports inside images when appropriate.

11.2 Comparison

Technique	Advantage	Limitation
Caesar	Very simple	Easily broken
Monoalphabetic	Large key space	Frequency analysis
Vigenère	Changes letter frequencies	Repeated-key attacks
Hill	Matrix-based encryption	Known-plaintext attack
Transposition	Rearranges letters	Preserves frequencies
Bell-LaPadula	Controls access	No integrity guarantee

11.3 Key Distribution

The network has 25 satellite clinics.

- Each clinic should have a separate key.
- Keys should be changed regularly.
- Keys should not be shared between unauthorized staff.

- A central administrator should manage key distribution.
- Staff should receive only the keys required for their work.

11.4 Limitations

Classical ciphers cannot guarantee:

- Strong confidentiality against modern attackers.
- Message integrity.
- Reliable doctor authentication.
- Non-repudiation.
- Protection against all insider attacks.

11.5 Final Recommendation

The combination of encryption and Bell-LaPadula provides basic protection in an educational model.

However, classical ciphers are not suitable for real patient records. Real healthcare systems require modern authenticated encryption, digital signatures, secure key management and proper access control.

Conclusion: A layered design is better than using a single classical cipher, but modern cryptographic techniques are necessary for real-world healthcare security.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design (8 Marks)

The new situation introduces an insider who has copied hundreds of encrypted messages using the same key. Messages also contain predictable headers such as PATIENT ID and DIAGNOSIS.

A. Impact Analysis (2 Marks)

1. Changes in Threat Model

Previously, the main concern was an external attacker.

Now, an insider at a satellite clinic has access to ciphertexts and can collect many encrypted messages.

2. Failed Assumptions

- One key is not safe for all clinics.
- Reusing the same key increases attack opportunities.
- Predictable message formats help attackers.
- Staff members may misuse their authorized access.

3. Increased Risk

Attack	Earlier risk	New risk
Eavesdropping	High	Very High
Frequency analysis	Medium	Very High
Known-plaintext attack	Medium	Very High
Key theft	Medium	Very High
Prescription modification	High	High
Replay attack	High	High

The attacker now has many ciphertexts and predictable plaintext formats, making cryptanalysis easier.

B. Evaluation of Original Design (2 Marks)

1. Vigenère Cipher

Vigenère uses a repeated keyword.

For example:

Plaintext: PATIENT

Key: MEDICAL

Ciphertext: BEWQGNE

If the same key is reused for many messages, repeated patterns can be studied. Kasiski examination can help identify the keyword length. After that, frequency analysis can be applied.

2. Monoalphabetic Cipher

The same plaintext letter always becomes the same ciphertext letter. Therefore, collecting hundreds of messages makes frequency analysis easier.

3. Hill Cipher

Hill cipher uses matrix multiplication. If enough independent plaintext-ciphertext pairs are known, the attacker may recover the key matrix.

4. Bell-LaPadula

Bell-LaPadula still controls which records the insider can read and write.

However, if the insider already possesses an authorized encryption key, Bell-LaPadula cannot prevent the insider from misusing copied ciphertexts.

Conclusion: The original design is not sufficient against an insider with many ciphertexts and repeated keys.

C. Adaptation or Redesign (2 Marks)

I recommend modifying the original design.

1. Use Separate Clinic Keys

Each of the 25 clinics should have a different key.

This prevents one stolen key from affecting the entire network.

2. Key Rotation

Change keys regularly to reduce the amount of ciphertext protected by one key.

3. Randomized Padding

Add varying padding to reduce predictable message patterns.

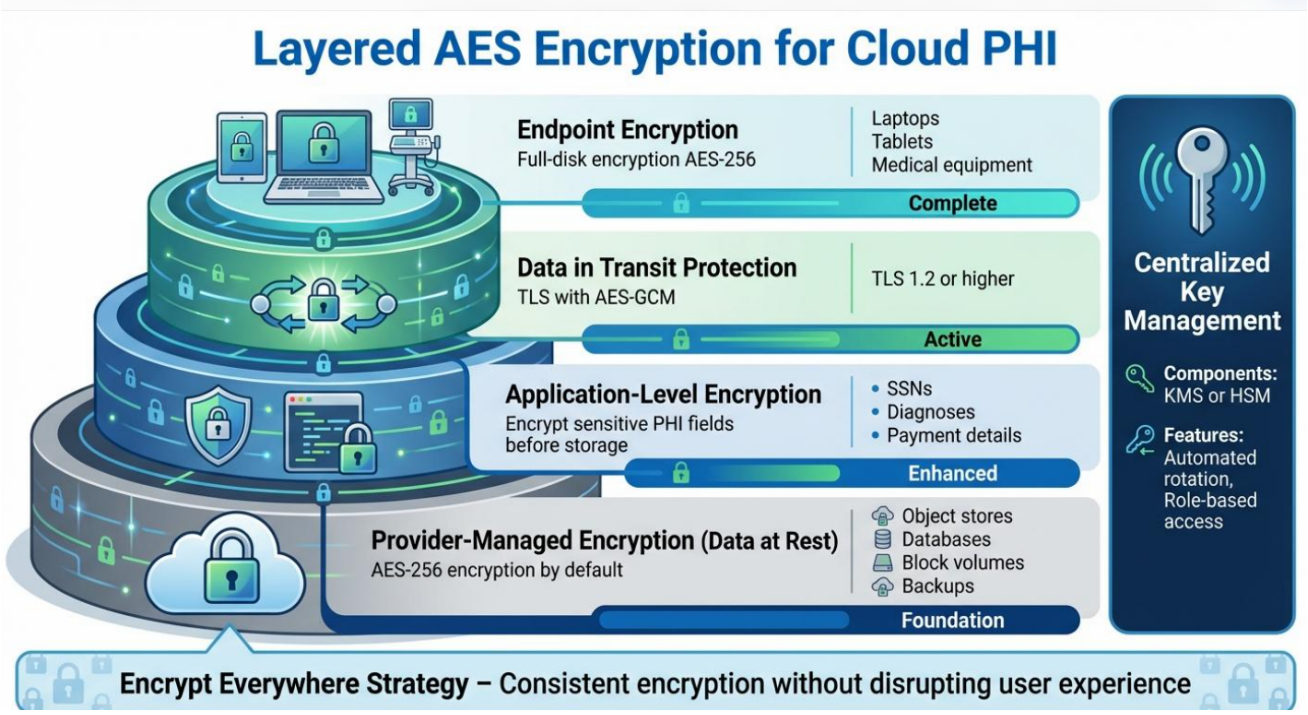
4. Remove Fixed Headers

Avoid repeatedly encrypting messages with identical structures such as PATIENT ID and DIAGNOSIS.

5. Use Encrypt-Then-Embed

Encrypt the short report first and then hide it inside a medical scan image.

Proposed New Design



New process:

1. Patient record is created at a clinic.
2. A separate clinic key is selected.
3. The record is encrypted.
4. Randomized padding is added.
5. The encrypted report is transmitted.
6. Bell-LaPadula checks user access.
7. The central hospital decrypts the received record.

Effect on Key Space: Separate keys increase the number of independent keys managed by the network. Key rotation reduces repeated use of the same key. However, increasing key space alone does not fix weaknesses in classical ciphers.

D. Evidence-Based Defence (2 Marks)

The modified design is better because:

- Separate keys reduce the impact of key theft.
- Key rotation limits ciphertext collection under one key.
- Randomized padding reduces predictable message patterns.
- Bell-LaPadula restricts unauthorized access.
- Encrypt-then-embed hides the existence of the report.

Limitations:

Classical ciphers still cannot guarantee strong confidentiality, integrity, authentication or non-repudiation.

Final Conclusion: The original design should be modified to use separate clinic keys, regular key rotation, randomized padding and layered encryption. For actual healthcare deployment, modern authenticated encryption and digital signatures must replace classical ciphers.

MARK DISTRIBUTION

Task	Marks
Task 1 – Threats, Attacks and Risk Evaluation	4
Task 2 – Security Goals Mapping	3
Task 3 – Bell-LaPadula Application and Critique	5
Task 4 – Cipher Design	4
Task 5 – Mathematical Analysis	5
Task 6 – Cryptanalytic Resistance	4
Task 7 – Transposition / Product Ciphers	3
Task 8 – Steganography Evaluation	3
Task 9 – Implementation & Testing	4
Task 10 – Experimental Evidence	3
Task 11 – Final Design Recommendation	4
Subtotal	**42**
Task 12 – Day-4 Design Challenge	**8**
TOTAL	**50**